

argiopeSegmentR

Masks the *opisthosoma* of *Argiope* with a trained U-Net and reads its dominant palette in CIELAB. The colour, artefact and QA stages are ported from SegmentR (Boyko 2025); the segmenter is the project's own.

v0.1.0 · R ≥ 4.1
unet resnet50 · 512 px
2026-09-04

INSTALLING

```
# once, from the tarball
install.packages("argiopeSegmentR_0.1.0.tar.gz",
  repos = NULL, type = "source")

library(argiopeSegmentR)
argiope_status() # is the machine ready?
```

The model runs in **Python**. If none is reachable, build one from R:

```
# ~500 MB, CPU-only torch, which is plenty
argiope_install_python()
# or point it at one you already have:
argiope_use_python("C:/.../python.exe")
```

THE FLOW, IN 9 STEPS

- 1 **argiope_status()** — check before you run anything
- 2 **argiope_gallery(dir)** — segments the folder, loading the model once
- 3 **argiope_items(g)** — the table: mask yes/no, score, size
- 4 **subset(it, !has_mask)** — what failed, and why
- 5 **argiope_palette_of(g, img)** — HEX + Lab + coverage
- 6 **argiope_plot(g, page)** — the grid, page by page
- 7 **argiope_dashboard(g, img)** — the four-panel dashboard
- 8 **argiope_card_grid(g)** — one card per specimen, colours named
- 9 **argiope_pdf(g, file)** — every page into one PDF

MINIMAL USE

```
g <- argiope_gallery("my_photos",
  out = "runs",
  run_id = "try1",
  n = 50) # sample

g
#> 50 images · 41 with a mask · 9 empty
#> median score 0.958 · pages of 6: 7

it <- argiope_items(g)
argiope_palette_of(g, it$image[1])
argiope_plot(g, page = 1)
argiope_dashboard(g, it$image[1])
```

DRAWING

argiope_plot	paginated grid; page, per_page, ncol, select, include_empty
argiope_pages	how many pages there are (0 if nothing to draw)
argiope_dashboard	four panels for one specimen; file= for a PNG. Returns the numbers
argiope_pdf	every page into one PDF
argiope_card	one specimen as a card: cut-out + palette, colours named
argiope_card_grid	a page of cards; reuses argiope_card
argiope_pick	multiple-selection list
argiope_pick_one	select a single one

SELECTING

```
# from a list (interactive)
sel <- argiope_pick(g)
# by name
sel <- it$image[it$has_mask][1:3]
# by position
sel <- 1:6

argiope_plot(g, select = sel, per_page = 3)
argiope_pdf(g, "sel.pdf", select = sel)
```

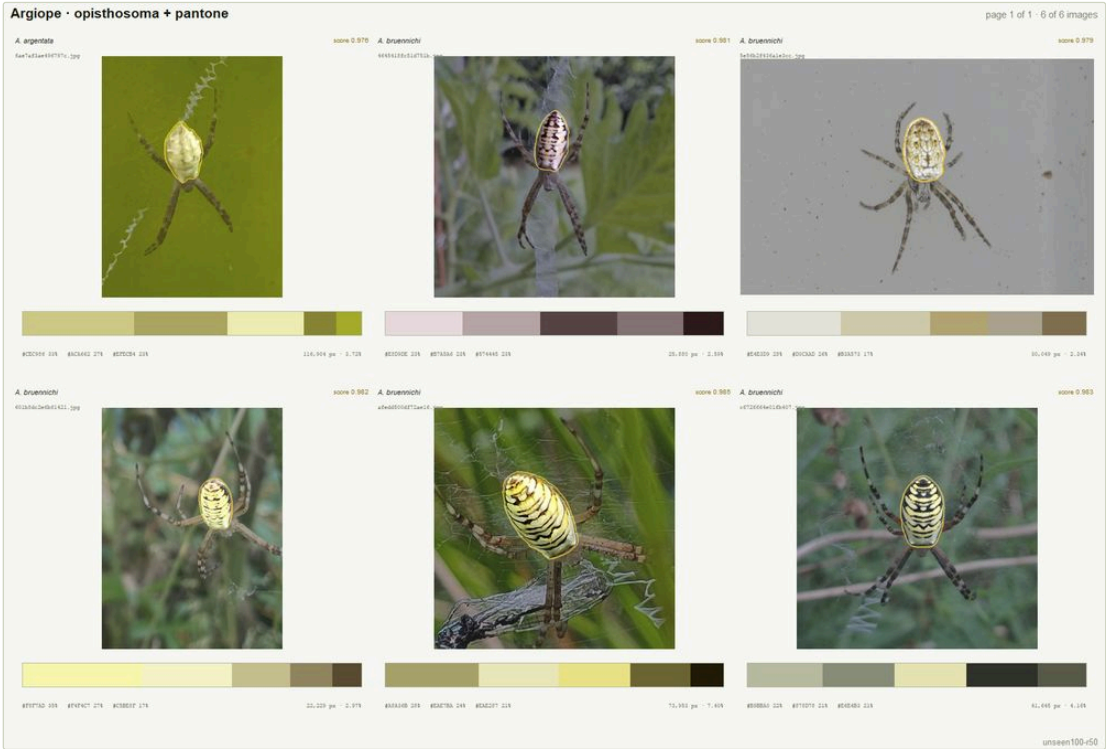
WHAT IT LEAVES ON DISK

run_config.json	every parameter: the run is reproducible from it alone
colors.csv	one row per image × cluster
skipped.csv	the ones that produced no mask, with the reason
json/*.json	per-image artefact (label, score, box, mask)
json/masks/	the binary masks, as PNG

Colour can be recomputed without loading the model:

```
python adapt_unet.py --from-json <run_dir>
```

What it produces



argiope_plot(g, select = ..., ncol = 3) — the photograph with the mask outlined in yellow and the background dimmed, the palette bar proportional to coverage, the leading HEX values, the score and the mask area. Six specimens checked by eye, from the pass over 100 unseen images.

How every segmenter tried has performed

SEGMENTER		MEDIAN IOU	MEAN IOU	IOU ≥ 0.7	EMPTY
GroundedSAM	SegmentR as published	produces no part mask at all — see below			
SAM 3, text prompt	project record	0.00	0.27	—	—
U-Net	resnet34	0.765	0.560	18/30	7/30
U-Net	resnet50	0.868	0.684	23/30	5/30

Same annotated held-out split (n = 30, the project's own partition, seed 42). Both U-Net rows were measured here with the same script; the SAM 3 row comes from the project record and has not been reproduced.

COLOUR SURVIVES THE MASK ERROR

Palette taken from the hand-drawn mask against palette taken from the predicted one, where IoU ≥ 0.7:

weighted ΔE (resnet50)	0.86
weighted ΔE (resnet34)	1.51
just-noticeable difference	2.3

Below the threshold: colour read off the automatic mask is **indistinguishable by eye** from colour read off the hand-drawn one.

ON IMAGES IT HAS NEVER SEEN

100 unseen GBIF	81 with a mask · 19 empty
40 annotation crops	39/40 · score 0.978
40 random ArTaxOr+GBIF	34/40 · score 0.951

Three failure modes, and which of them a filter can catch

1 · NO MASK

Roughly **one field photograph in five** comes back with nothing. It is recorded in `skipped.csv` with its reason, never dropped in silence.

Detectable: yes — `has_mask == FALSE`.

2 · FRAGMENT

3 of 81 masks covered less than 0.1 % of the frame: scraps, not abdomens. All three scored **below 0.70** against a median of 0.959.

Detectable: yes — filter on score, or on area.

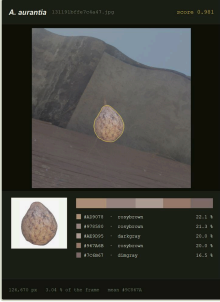
3 · EGG SAC

The model segments **egg sacs** as though they were abdomens, and scores them like good ones. **Three of the twelve highest-scoring** masks over the 100 unseen images are egg sacs. Neither score nor area catches them: they are large and compact.

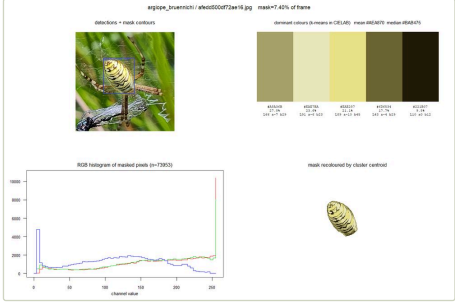
Detectable: not automatically. Someone has to look.



`arglope_card_grid(g, select = ..., ncol = 3)` — one card per specimen: the opisthosoma cut out beside its palette, every colour named by the nearest R colour, computed in CIELAB.



`131191bffe7c4a47.jpg` — there is no spider in the frame. The mask encloses an egg sac stuck to a wall, at score 0.981, and the "opisthosoma palette" is five shades of the sac's papery surface.



`arglope_dashboard(g, img)` — detection and contour · dominant colours with HEX, coverage and Lab · RGB histogram of the masked pixels · the mask recoloured by assigning every pixel to its centroid.

Colour, artefact and QA stages ported from **SegmentR** — Boyko, J. D. (2025), *Ecological Informatics* 90:103259, MIT licence. The segmenter and the R layer belong to the Arglope project. Colours are reported as HEX, Lab coordinates and coverage; no PANTONE® compatibility is claimed. The photographs are third-party observations via GBIF, most of them CC-BY-NC — no commercial use.